

Introduction to AWS

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What is Amazon Web Services(AWS)

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Amazon Web Services offers a broad set of global cloud-based products including compute, storage, databases, analytics, networking, mobile, developer tools, management tools, IoT, security, and enterprise applications: on-demand, available in seconds, with pay-as-you-go pricing.

What is AWS ?

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, offering over 200 fully featured services from data centers globally.

Region and Availability Zones

Region - separate geographic area

Availability Zones - isolated locations/ data centers within each Region



AWS Global Data Centers

- **Geographic Regions- 31**
- **Availability Zone - 99**
- **Edge Locations - 410+**
- **Countries- 245**
- **Direct Connect Locations- 115**



**Coming soon 5-AWS Region and 15 more
Availability Zones**

Understanding AWS Free Tier

Types Of Offers:

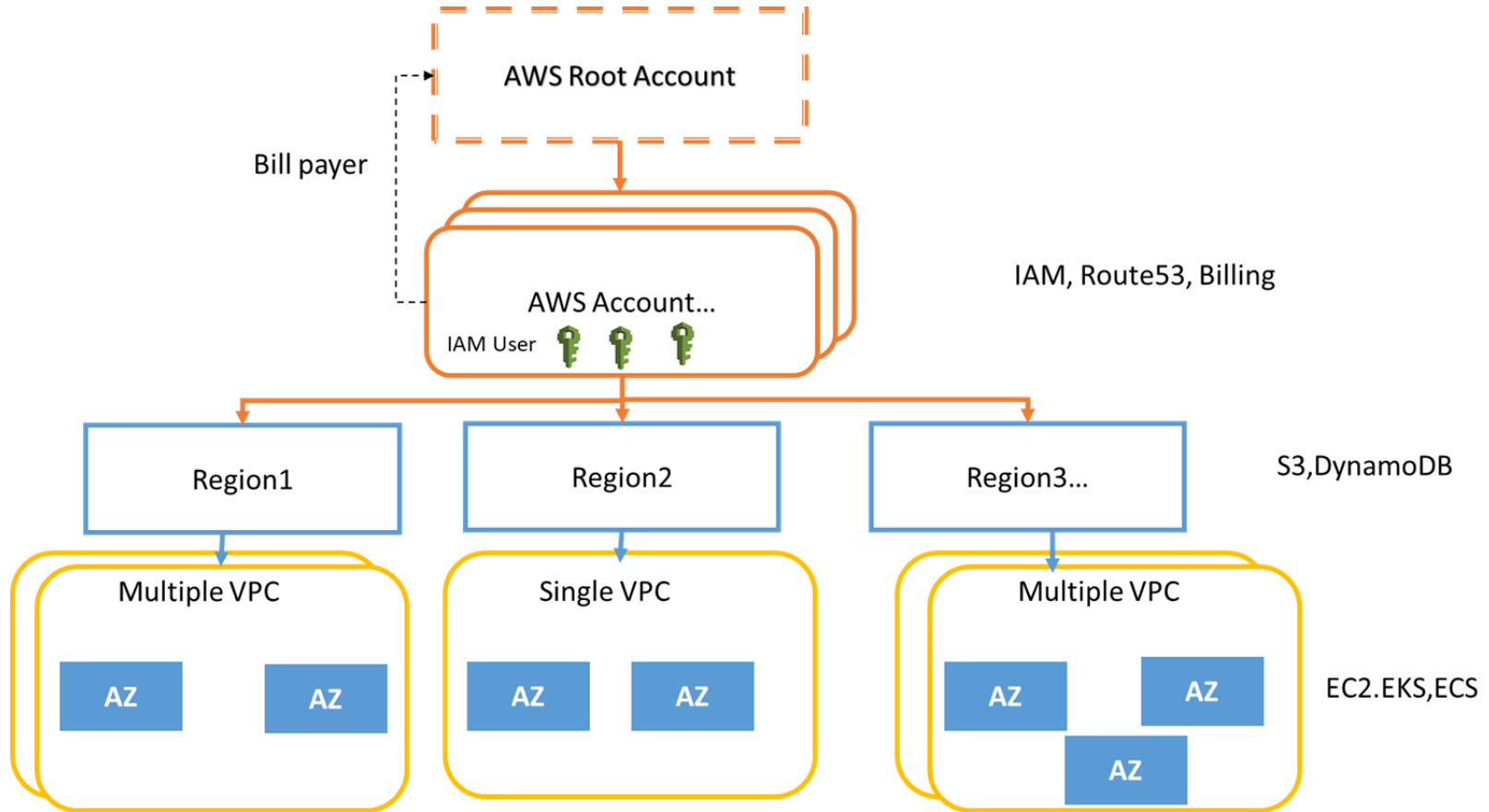
1. 12 month Free:

- EC2 & ELB - 750 Hour, ECR - 500 MB
- S3 & EFS - 5GB, CloudFront - 1 TB, ELB- 30GB, ECS - 2200 Hours
- RDS,ElastiCache - 750 Hours, DynamoDB- 25GB,Redshift- 2 Month

2. Always Free: Lambda,SQS & SNS - 1 million,DynamoDB- 25GB, CloudWatch - 10,SES- 6200 messages/month

3. Trials: Sagemaker & Redshift- 2 months, Appstream 2.0- 40 Hours

AWS Organizations and account structure:



Configure AWS IAM User

1. Create IAM admin user

- IAM Service -> Users -> Add User
 - Access Type: Programmatic Access, AWS Management Console
 - Console password: Custom password
 - Select - Users must create a new password at next sign-in : Yes
 - Next Permissions: Attach Existing Policies directly -> Select "Administrator Access"
 - Review and create User
- Save credentials
 - AWS Login URL
 - IAM username/password
 - IAM Access key/Secret Access key pair
 - Download CSV
- Logout from current root user session and Login back using sign-in link and IAM username/password that you find in the downloaded CSV/Excel

Set Billing Alarm for your AWS account

1. Set Billing Alarm

- Login to AWS account using Email/Password
- Billing Dashboard -> Preferences -> Receive Billing Alerts
- Create Billing Alarm
 - CloudWatch -> Select N.Virginia region
 - Alarms -> Billing -> Create Billing Alarm

2. Grant IAM users permissions for Billing and Cost management

- My Account -> IAM User and Role Access to Billing Information -> Activate

Understanding about AWS services

AWS Services

Deployment & Management

Application Services



Amazon SQS



Amazon ElasticTranscoder



Amazon SES



Amazon AppStream



Amazon CloudSearch

Mobile Services



Amazon Cognito



Amazon Mobile Analytics



Amazon SNS

Enterprise Applications



Amazon WorkDocs



Amazon WorkSpaces



Amazon WorkMail

Application Services

Administration & Security



AWS DirectoryService



AWS IAM



AWS Trusted Advisor



AWS Config



AWS CloudTrail



AWS CloudWatch

Deployment & Management



Amazon CloudFormation



AWS OpsWorks



AWS CodeDeploy

Analytics



Amazon Kinesis



AWS Data Pipeline



Amazon EMR

Foundation Services

Compute



Amazon EC2



AWS Lambda

Storage & Content Delivery



Amazon CloudFront



Amazon Glacier



AWS Storage Gateway



Amazon Content Delivery

Database



Amazon DynamoDB



Amazon RDS



Amazon Redshift



Amazon Elastic Cache

Networking



Amazon Route 53



Amazon VPC



AWS Direct Connect

Launch EC2 Instance

Initiate instance launch [link](#)

Step 1: Open the Amazon EC2 console at <https://console.aws.amazon.com/ec2/>

Step 2: From the Amazon EC2 console dashboard, choose Launch instance.

Step 3: Choose an Amazon Machine Image (AMI)

Step 4: Choose an Instance Type

Step 5: Configure Instance Details

Step 6: Add Storage

Step 7: Add Tags

Step 8: Configure Security Group

Step 9: Review Instance Launch and Select Key Pair

Thank You